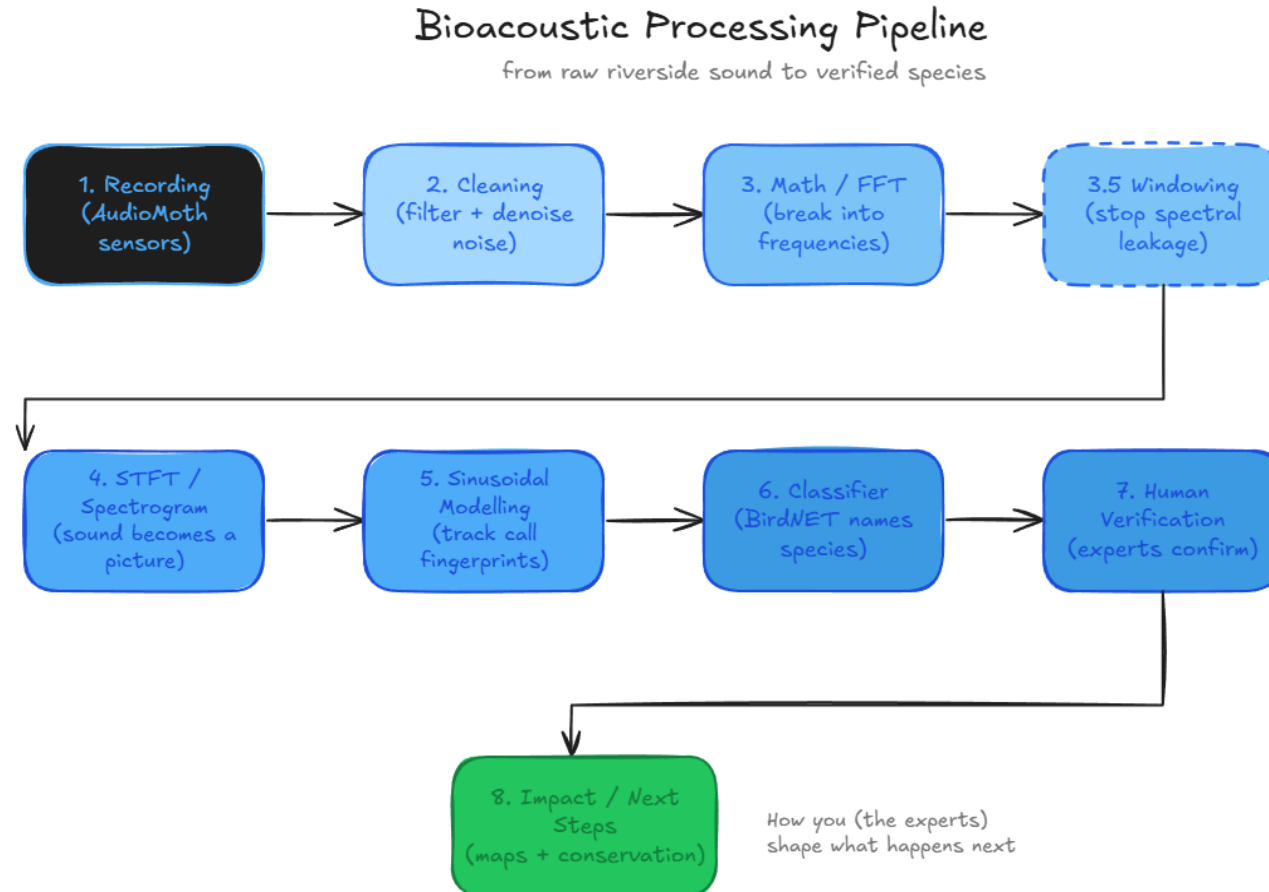


What Is The Problem? + Pipeline Overview

"What is the spatial-temporal distribution of bird species along the Tyne riverside ?"

- Autonomous recording units (**ARU**) in the riverside capture enormous amounts of data, composed of environmental sounds, bird calls, human activity, and traffic.



Processing can be automated, but verification still requires human verification.

The Recording Process

- Audio moths capture 14 hours of continuous bird data , capturing up to **24Khz** and sampling at **48Khz**.
- It processes thousands of hours of audio data which would take incredible amounts of time for us humans to process, hence why an automated pipeline is crucial to make the finding of bird species feasible.
 - One morning recording is **2.3GB** of data; across all 10 sites it adds up to **200-330 GB+** per week.
- Nyquist theory: Sound must be captured at 2x of the highest frequency, or aliasing can be introduced.
 - Wetland, woodland, and residential areas (birds close to human activity vs natural habitats)

1. RECORDING — sensors listening along the river



hours of
audio data

approx. 2.25 GB
per morning +
2.25GB per night,
per site

Audio Cleaning (Filtering+ Denoising)

- Before we do any processing work, we must clean the recording from everything that obfuscates bird frequencies; especially at low frequencies as human activity, wind, riverside rushing water, and other factors affect the quality of results.
- Research on riverside sites in Italy have shown that close traffic and river noise masks crucial bird activity especially at low frequency)
- To tackle this, we use 2 techniques **band-pass filters** and **wavelet denoising**, to diminish the unwanted sounds.
 - Band-pass filter allows only specific ranges of bird frequencies through while cutting the frequencies above and below that range
 - Wavelet denoising allows high-frequency bird sounds to shine through, while ensuring environmental noise does not diminish/drown out the signal for us or the computer machines (this is a mandatory pre-processing step, if the audio remains uncleaned the automated "birdwatcher" has no direction; running blind)
 - Wavelet denoise can also persevere structured low-frequency calls (e.g. owl), while still removing low-frequency environmental noise (e.g. wind), as it separates sounds by their structure across various levels and not just the frequency they are in.

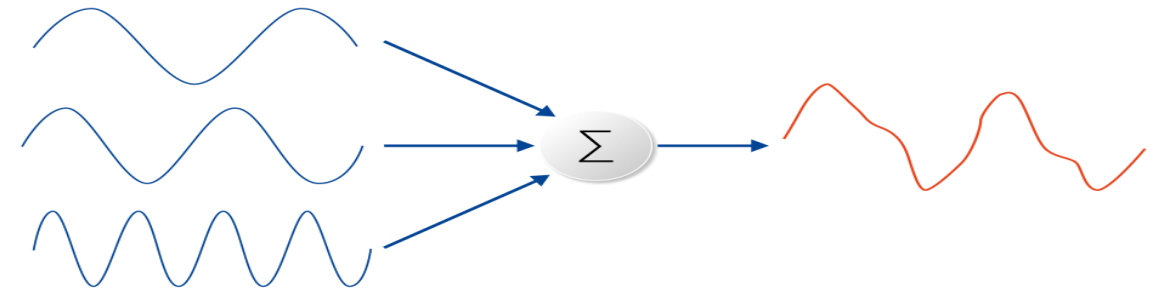
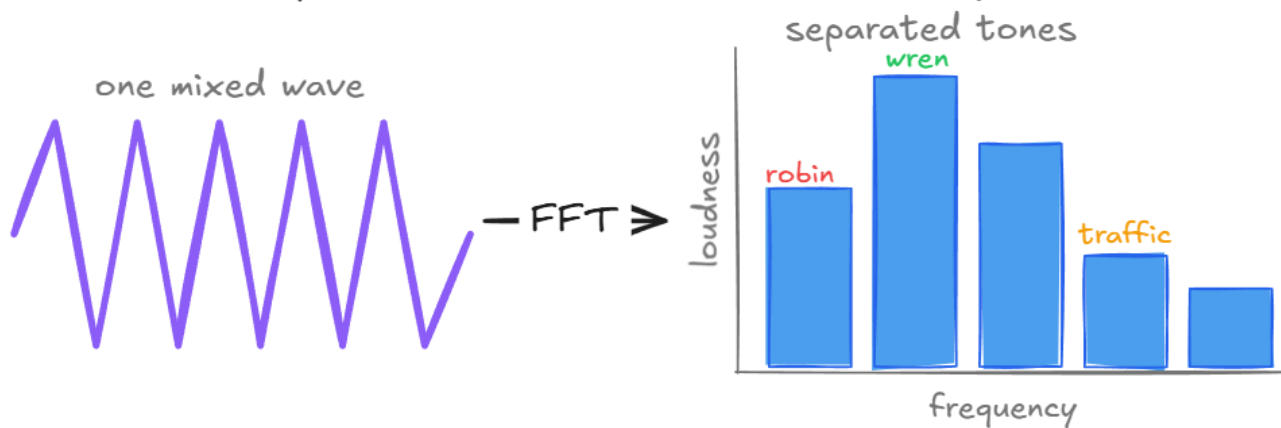
2. CLEANING — strips the noise, keep the bird calls



The Math Used (FFT)

- Every sound we have ever heard is a mix of frequencies bundled together just like ingredients in a recipe.
- We use a fast algorithm called **FFT** (Fast Fourier Transform) to ask the audio which frequencies are present and how much ? , it pulls mixed waves into separate identifiable tones, which have their set of combined proprieties which make it unique (amplitude, frequency, phase).
- FFT is the improved algorithm deriving from the older DFT allowing us to turns months of computation into minutes in the fraction of the time, which is why processing **200-330GB** of riverside data per week is even feasible.
- Eulers formula is the math that has been around since the **18th century** which makes all this possible, it can break down any sound wave (no matter how complex) into a combination of simple rotating cycles, the FFT essentially does that.

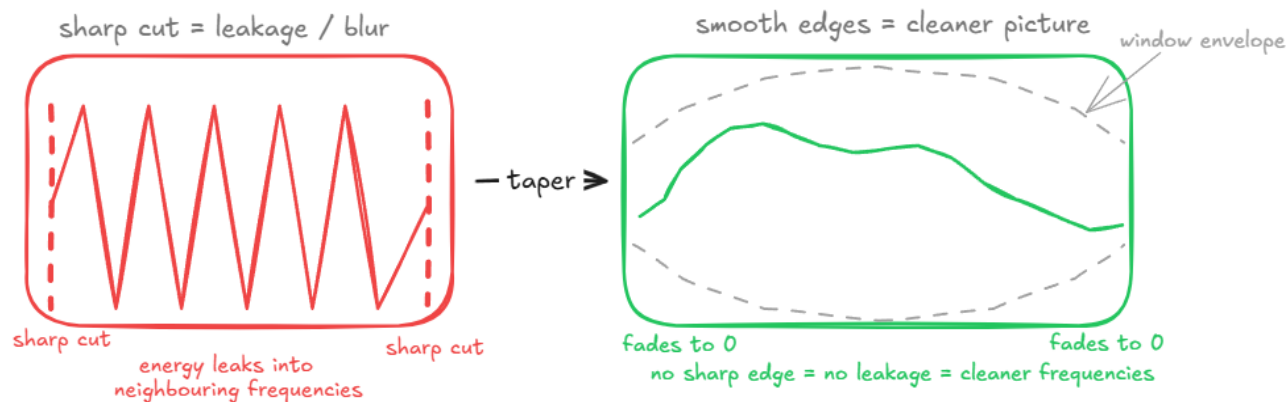
3. FFT, splits the sound into its frequencies



Why windowing is necessary?

- To analyse long recordings, we must first cut it in shorter chunks, however cutting at sharp edges introduces a problem called spectral leakage, this is when energy from a frequency smears into its neighbours blurring the actual picture of the sound, potentially burying the faint call of other birds under a loud .
- Windowing functions like Hann, Hamming or Blackman are used to taper the edges of each block smoothly, each making a slightly different trade-off between preserving frequency detail and suppressing leakage, hence why choosing the right windowing function is important.
- The window size we choose directly affect time and frequency resolutions (Trade off):
 - Short Window provides high time resolution meaning it can capture rapid, sharp bird calls clearly but with a coarser frequency.
 - Long Window provides high frequency resolution, meaning it can clearly separate 2 similar pitches but blurs fast changes in exchange.
- BirdNET analyses audio in 3 second sliding windows, as the model was trained on 3 second clips which capture most bird calls completely, duration matters (too short you miss bird calls, too long we lose accuracy)

3.5 WINDOWING, we must taper the edges so the picture isn't blurred



The trade-off
you must choose:

Short window
= high time resolution
catches fast bird calls

VS

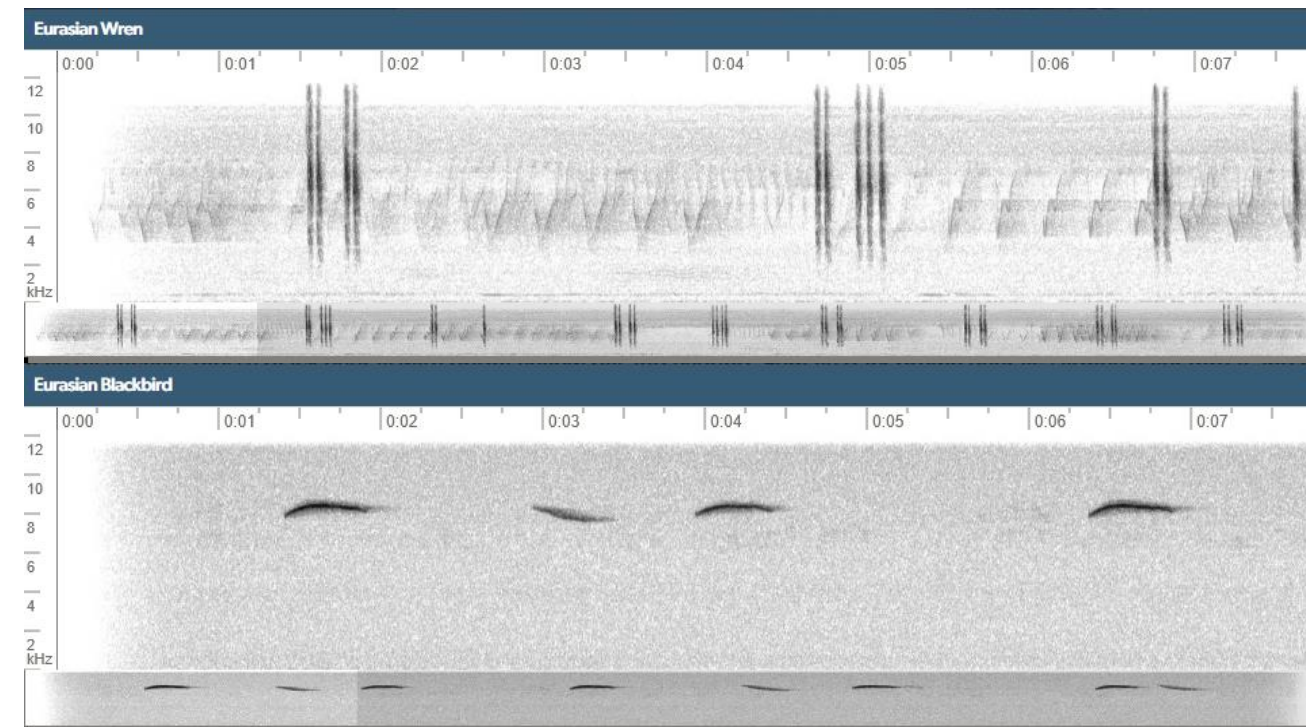
Long window
= high freq resolution
separates close pitches

BirdNET research: short window wins for birds, as calls are fast and sharp.
Windowing fixes leakage, filtering removes noise, Two separate jobs.

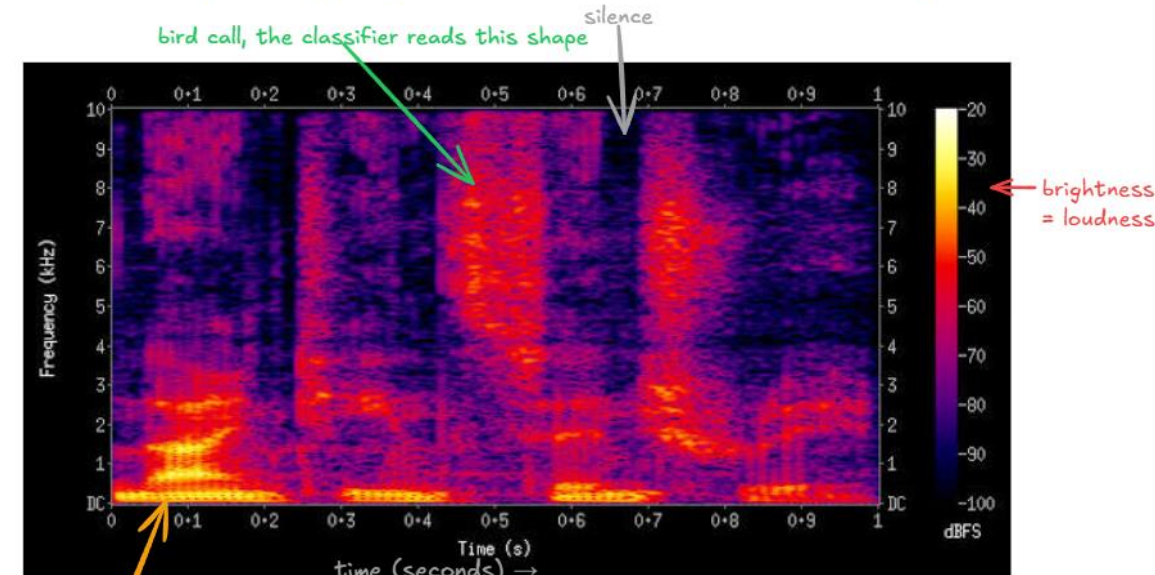
STFT & Spectrogram

- The Spectrogram is a picture of the sound where time runs on the horizontal- axis, frequency (pitch) on vertical axis and the loudness of each frequency at each moment is represented by brightness.
- The SFTF (Short-Time Fourier Transform) is exactly what produces this image by sliding the analysis window across the entire recording, taking an FFT at each step and layering them side by side.
- Every modern bird classifier uses this image (spectrogram), rather than the raw audio file itself. Every step before this slide exists to produce the best spectrogram possible, while everything after exists to read/process it (The windowing step earlier directly shapes this image).
- BirdNET's research reveals that short-window (high time resolution) works best for birds, as their calls are so rapid and sharp.

the call is teh fingerprint of a bird, one is melodic while the other has rap



4. STFT / Spectrogram, sound when it's turned into a picture



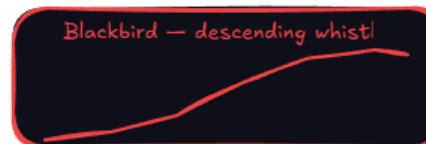
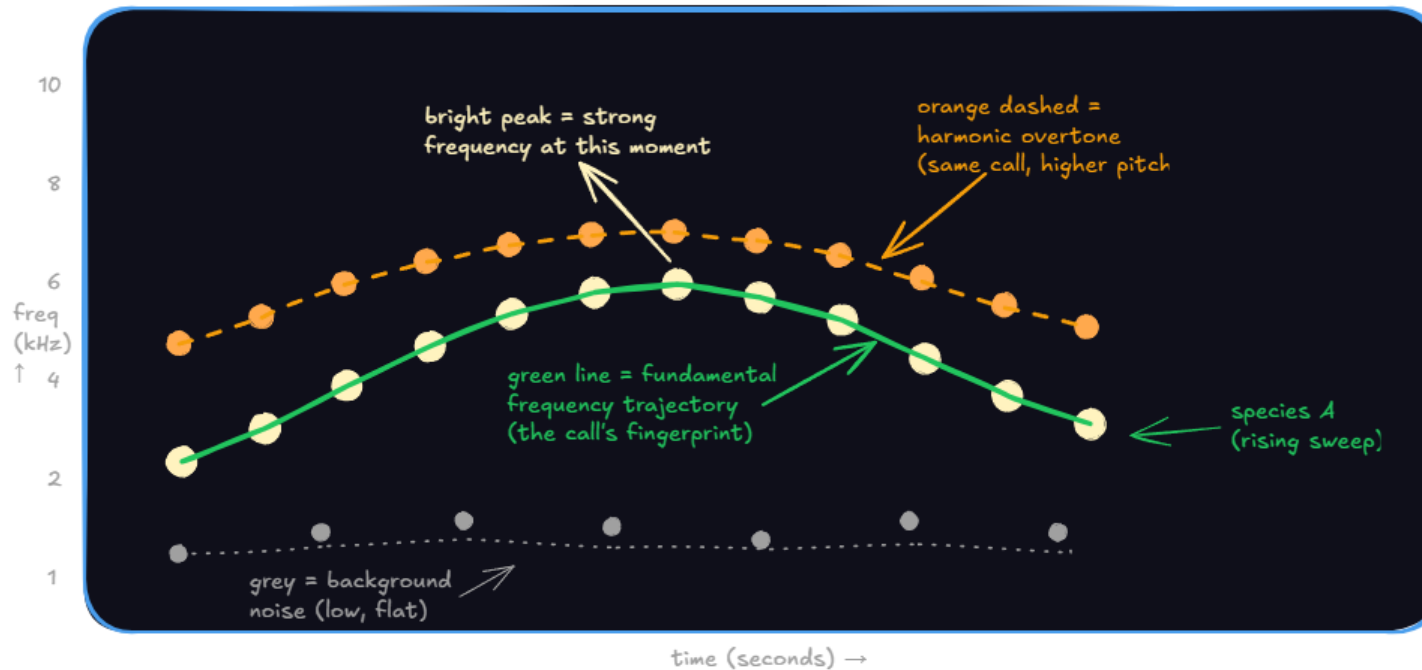
Every modern bird classifier looks at this picture, and not the raw audio.

low-freq background noise (filtered out)

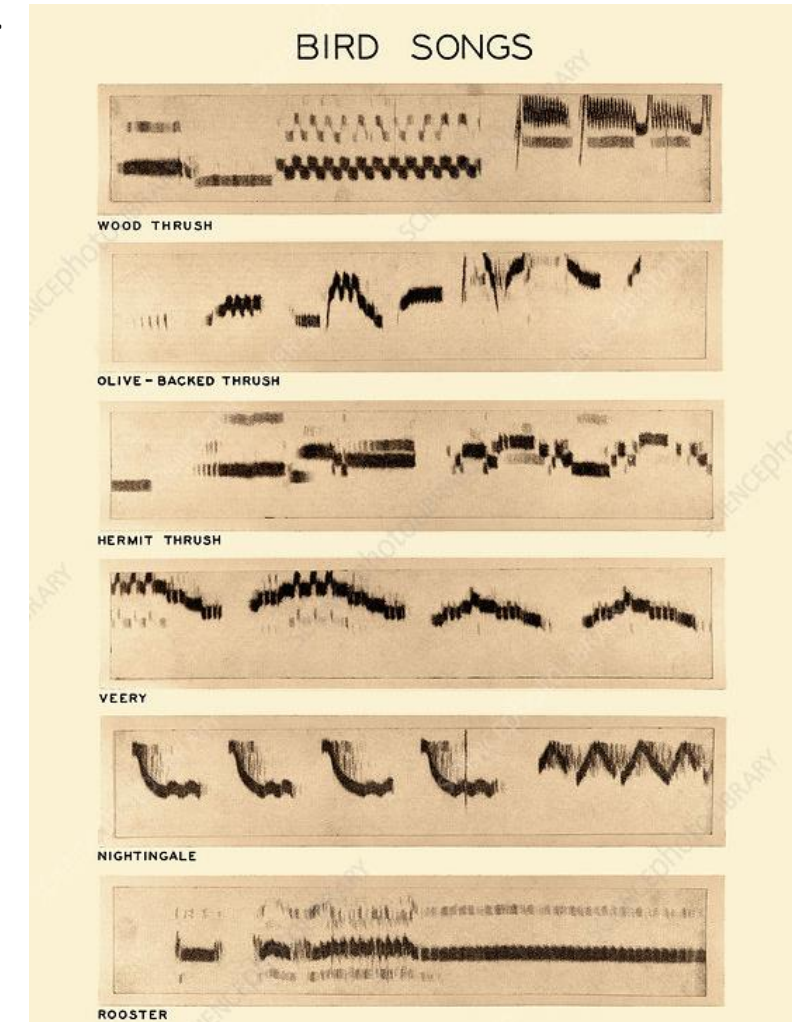
Sinusoidal Modelling

- Sinusoidal modelling consists in find the brightest peaks of energy (strongest frequencies at each instant) and connect them overtime into trajectories.
- Outlining these peaks makes it much simpler for human eyes and machines to visualise the individuality of a robin's call compared to a balckbird, as each have their own unique characteristically patterns (composed of descending whistles, rising sweeps, trills, and rising sweeps), a kind of biometric signature.

5. Sinusoidal Modelling, when we trace the call's fingerprint



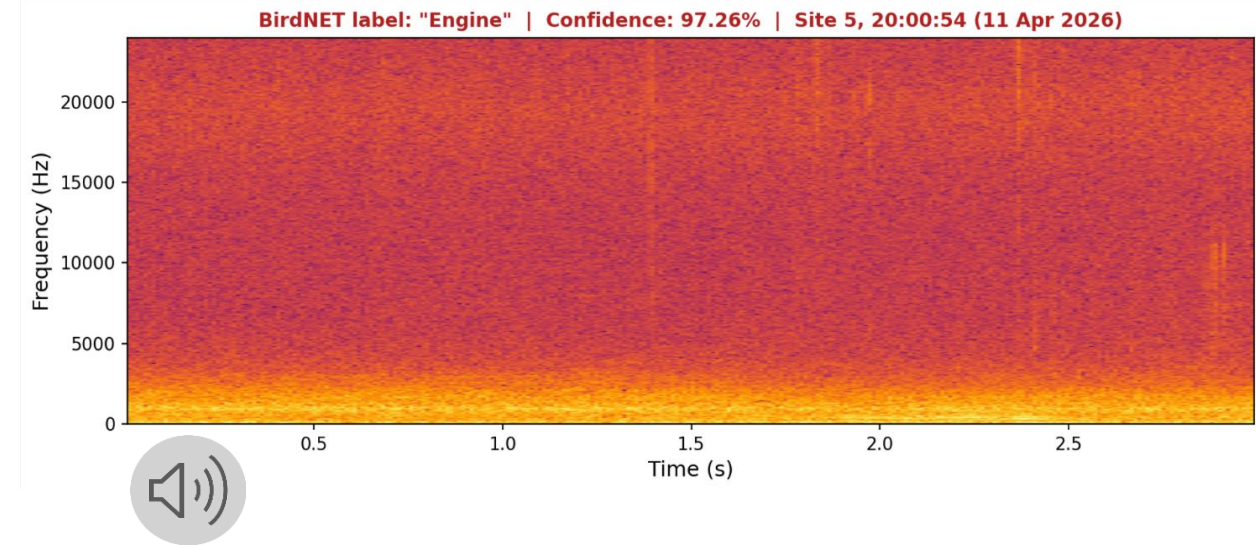
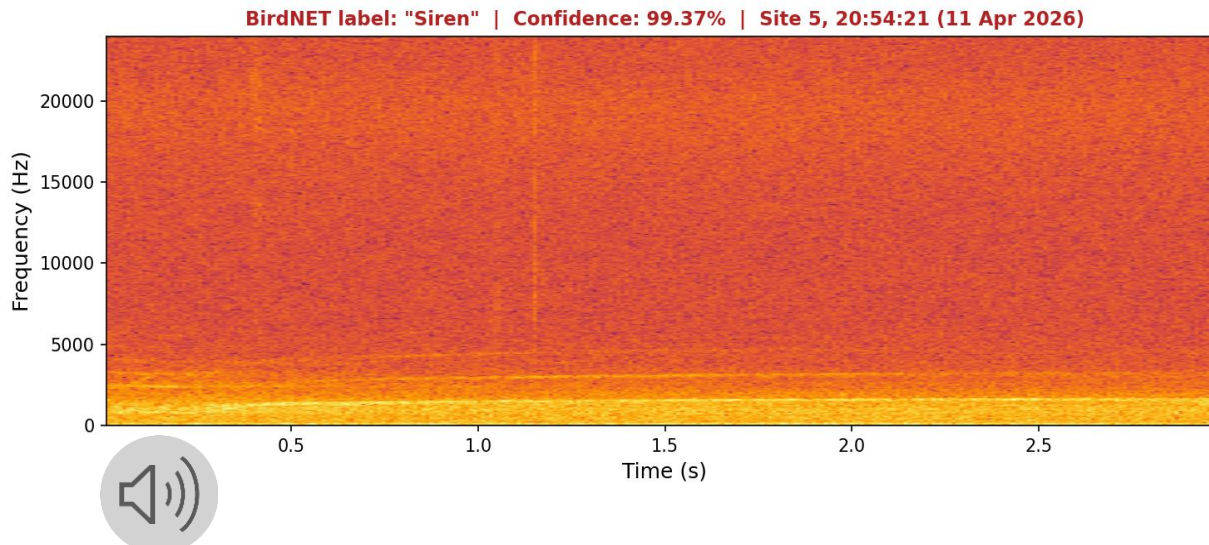
Each species leaves a distinct shape, and that's the fingerprint the classifier learns to recognise.



BirdNET Classifier

- Once we have a clean picture (spectrogram), we can have the ML (machine learning) "birdwatcher" analyse it.
- ML models like BirdNET don't listen the audio itself but rather analyse the spectrogram images using computer vision techniques (same technology used for facial recognition), to predict which bird species are singing.
- BirdNET treats spectrograms as monochrome image, and it can identify up to 6,500+ species simply by the shape of their sound.

6. CLASSIFIER, BirdNET model

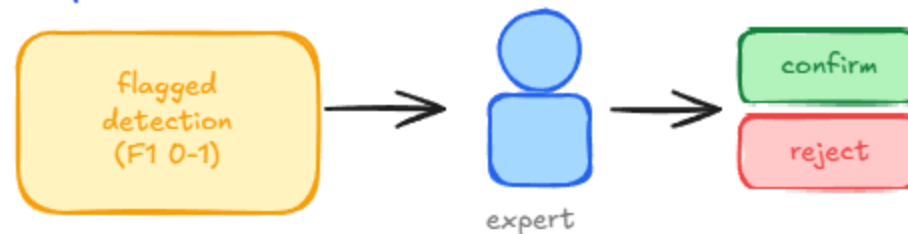


No clear bird call shape/fingerprint just uniform background noise spread across all frequencies flagged as 97%+ accuracy, hence why we need expert verification.

Why Machines Still Need Humans?

- All the hard work is done by the automated pipeline (data sorting, the mathematical processes, making spectrograms and assigning confidence scores) , however this does not leave humans completely out of the equation.
- We still need bird experts to review the results, confirm uncertain results, find missing species (which bird experts can provide the lack of ecological context as to why that is).
- The pipeline's objective is to slim down the work experts must do, from process all the data and analyse to review and verify the detections made by the ML model, which is exactly what allows for more reliable and refined results at this extensive amount of data.
- A European study across 194 sites found that messy data recorded in the wildlife rather than a clean lab recording dropped the model's performance to an F1 score below 0.5
- An F1 score of 0.5 stands for when a model makes enough mistakes to miss bird calls and display false alarms; decent enough as a result but needing significant verification, while an F1 score of 0.8 on the other hand would stand for a solid result but still needing some verification.
- The model has been shown to make errors even in our own pipeline, where it detected Engine 596 times and Siren 24 times in one instance, however those are clearly false positives as there's no such bird species and it is most likely referring to car engines and ambulance/police sirens, this being concrete proof that we need human verification no matter how good the model we are using may be.

7. HUMAN VERIFICATION, we need the expert's verification



Impact & Next Steps

- Once experts verify the detections they can be turned into powerful analytical insights, helping us understand which species are where and when across the riverside in a map format, highlighting how patterns change from the beginning to end (e.g. missing species, bird calls only at specific times, more human activity), these patterns are exactly what helps us predict ecological change (change in migration timing, noise pollution, habitat loss).
- None of this is possible without the people in this room from experts to bird enthusiasts; the pipeline helps us understand what it heard, but only an expert can help us understand what that means and what things are missing and why. You have a role in this that no machine can replace, turning hours of riverside data into the crucial awareness needed to protect the birds living there.
- As we continue to analyse these recordings, we may be able to document species that we never thought could be living in the riverside, every detection being verified by experts, alongside unexplainable missing species will lead to a better understanding of how to best protect the riverside.

8. IMPACT, verified data drives
conservation across habitat

species maps

spot what's missing

protect habitat

